Volkswagen AG

Qualification Programme
New Parts

Procedures with integrated acceptance of 2-day production

1th revised edition

March 2006

1st version September 1991
New version, April 2003
1th revised edition, March 2006

Information concerning further and supplementary quality management components of the Volkswagen Group:

***Formula Q new parts – integral (*Software)**

Qualification Programme for New Parts

The software that belongs to this manual provides the option of handling the Qualification Programme for New Parts with PC support. It is the preferable way of working.
The results can be sent to the recipient per e-mail.

Download from http://www.vwgroupsupply.com

Formula Q - Konkret

Quality management agreements between the Volkswagen Group and its suppliers

Download from http://www.vwgroupsupply.com

Formula Q - Capability

Quality capability of the suppliers
(Evaluation guideline)

Download from http://www.vwgroupsupply.com

Quality Documentation for pre-series phase

Information on parts designation during the pre-series phase and guidelines for sampling.

Download from http://www.vwgroupsupply.com

Publisher: Volkswagen AG,

Corporate Quality Assurance, Purchasing

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Introduction

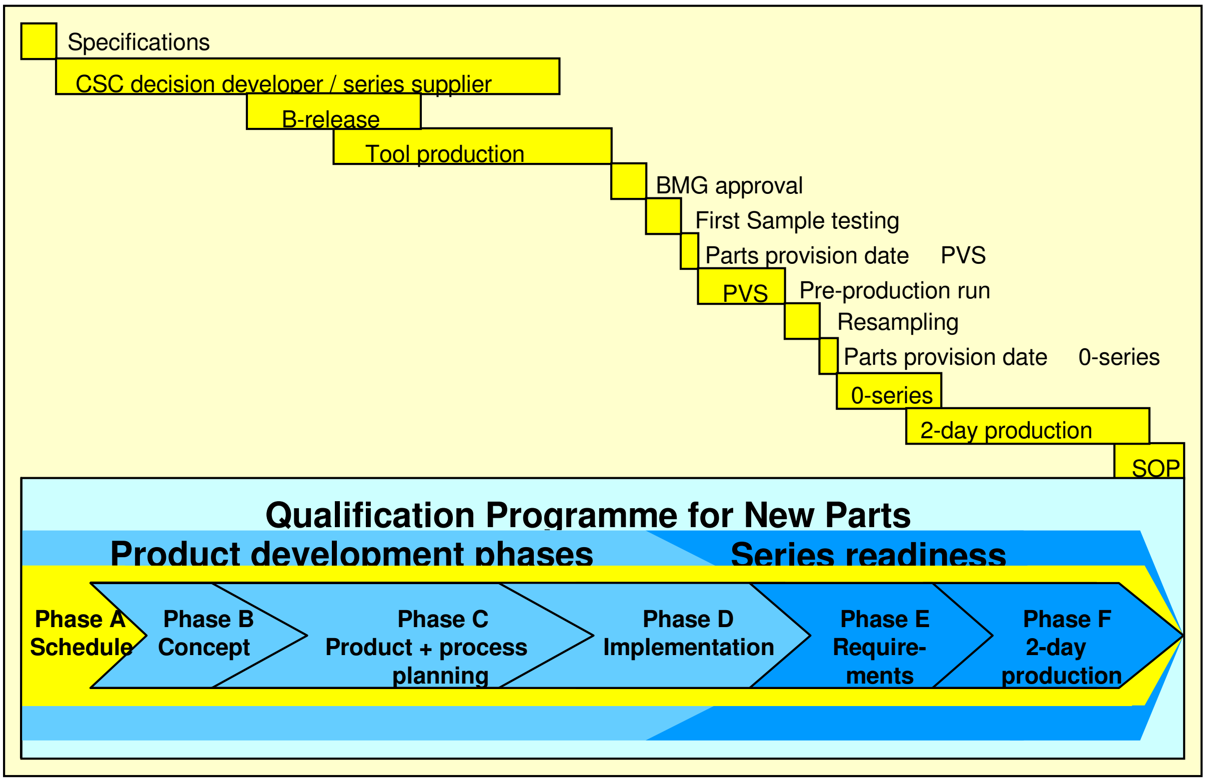
1.1 Purpose

The Qualification Programme for New Parts is a tracking system for purchased parts.

The main objective is to achieve the quality agreed upon and the amount of purchased parts for a specific vehicle project on time.

The Qualification Programme for New Parts is a guideline with a standardised procedure and will be used throughout the entire Volkswagen Group. It simplifies the cooperation between our suppliers and the plants that use their components for the first time.

The basis for the implementation of the Qualification Programme for New Parts (hereinafter referred to as QPN) is the quality management agreement, "Formula Q – Konkret", between the Volkswagen Group and its suppliers. The quality management agreement is a supplement to the purchasing conditions for production materials and constitutes an integral part of the delivery contract. The supplier has to start working on the QPN as soon as he receives an order.



To suit the project and part specific requirements, the Qualification Programme for New Parts was built in a modular way and structured according to the phases of product development and series readiness.

In the **product development** phases, the project progress of the purchased parts is determined. Individual parts with the same or a similar production process are allocated to part families (see 1.2) and their Q-status is evaluated with questionnaires / checklists.

In the **series readiness** phases, the supplier has to prove that he can produce the product on time, to the quality standard and in the quantities agreed upon.
The necessary requirements are checked with a checklist.

If the quality assurance department of the customer plant accepts that the component requirements are fulfilled, a 2-day production audit can take place. If it is successfully concluded, the Qualification Programme for New Parts has been completed.

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The quality assurance department of the plant that uses the product for the first time, hereinafter referred to as QA customer, provides a preliminary prioritisation of the order.

The prioritisation also defines the intensity of the cooperation between the QA customer and the supplier (Chapter 1.4 "Part-specific determination of the priority number).

1.2 Application area and handling information

The Qualification Programme for New Parts is used for all new parts in new projects (except standard parts) as well as for all new suppliers for existing projects.

The individual parts (part numbers) of a project scope can be combined and processed as a "parts family" in the Qualification Programme for New Parts to reduce the processing effort (e.g. protective side rail, grained, front left/right, rear left/right...)

Complex order scopes (many versions) can be combined to form a "parts family" if the complexity is based on a basic version and results from:

- Additional processing steps (e.g. painting, etc.)
- Assembly of different components of the respective versions

Example: Protective side rail, grained, 1K3 853 515 / ...
Protective side rail, painted, 1K3 853 515 A / ...
Protective side rail, painted, with decorative insert 1K3 853 515 X / ...

Requirements for the definition/handling of a parts family have the same deadlines for application and the same development, planning and implementation phases.

The numerically smallest part number of the order scope has to be entered as reference part number for a parts family that is processed/evaluated in the Qualification Programme for New Parts. For the example of the protective side rails, the reference part number is: 1K3 853 515 (without index)

Order scopes for which no specific part number can be defined for reasons concerning system or production control (usually modules or JIT orders), are always classified with Priority 1 – important parts with special monitoring.

For these scopes, the reference part number has to be determined in cooperation with the parts manager who acts as a contact person for the QA department of the plant that uses this product for the first time.

The processing of the Qualification Programme for New Parts always starts with Phase A, if no other arrangements were made with the QA department of the customer. Phase A includes the detailed coordination of all current and quality-relevant deadlines and project data.

In the software version of the QPN, the processing of Phase A also includes the standardised preparation (basic data capturing) for all evaluation documents used.

The processing of Phase B, i.e. the detailed evaluation of the concept with regard to quality-relevant aspects follows immediately thereafter. If this evaluation was already performed prior to the order, jointly with the QA department of the customer (e.g. within the framework of a concept tender), only updating of the information is required.

Phases C to F of the Qualification Programme for New Parts are then processed. Deadlines and quality-relevant project information have to be updated.

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If no contact person in the QA department has been appointed within the framework of the order placement, the identity of such a person has to be obtained from the contact person in the purchasing department.

The result of each evaluated project phase has to be sent per e-mail (in exceptional cases per fax) to the responsible staff member of the QA department in the plant that uses the product for the first time, without a special request being made.

If the supplier has not been given a priority number, the processing is always performed with Priority Number 3.

The responsible staff member of the QA department of the customer plant will contact the supplier after receipt of the First Result documentation for parts with priority 1 and 2 (see Chapter 1.4) to conduct a project meeting.

Introduction

Overview of the phases of the Qualification Programme for New Parts:

Phase	Evaluation phase	Deadlines for presentation of the evaluation results to the customer	Result documents
A*	Project information and deadlines	At the latest 3 weeks after placing of the order (receipt of the letter of commitment)	(1) , (2) , (3)
B*	Concept		
C*	Product and process planning	At the latest 6 weeks after placing of the order	(1) , (2) , (3)
D	Implementation	After positive evaluation of the product and process planning phase, at the latest 4 weeks before the pre-production run (PVS)	(1) , (2) , (3)
E	Requirements for acceptance of 2-day production (quick check)	After positive evaluation of the implementation phase, at the latest 1 week before the planned / agreed acceptance deadline for 2-day production (2DP)	(1) , (2) , (3)
F	Acceptance of 2-day production	After positive evaluation of phase E and the initial sample test, at the latest 2 weeks before the Start of Production (SOP)	(4)

* Evaluations may be combined, depending on the required value-adding process (development and/or production) and if possible within the required period.

Result documents expected by the customer:

(1) : Quality framework schedule (updated to current level)
(2) : Description of the planned production sequence (updated to current level)
(3) : Completed questionnaire (checklist) for the relevant evaluation phase with result documentation (result sheet, Chapter 5.1)
(4) : Acceptance protocol concerning 2-day production (Chapter 4.2)

Note: Within the framework of the contract, the supplier is obliged also to perform all the required qualification measures for his sub-suppliers (refer Formula Q – Konkret).

In this case we recommend to our suppliers to apply the current Qualification Programme for New Parts or a similar system to the sub-suppliers.

1.3 Evaluation procedure and documentation of results

The individual QPN phases are evaluated by answering the questions provided. If part-specific, additional questions are asked in the respective project phase, they must be included in the evaluation.

The evaluation is performed according to the following grading:

10	points	=	Requirements fully compliant.
8	points	=	Majority* of requirements compliant; minor deviations.
5	points	=	Some of the requirements compliant; serious deviations
0	points	=	Requirements not met; major deviations.
X		=	Question not relevant for this evaluation (Question should be temporarily or permanently excluded from the evaluation)

After the first question that was evaluated with 0 points or the third question that was evaluated with 5 points, the project status is set to YELLOW. It is mandatory to provide information concerning the corrective measures initiated, the implementation deadline and the person responsible for the implementation for requirements that were evaluated with 0 to 8 points. For requirements that were evaluated with 10 points, a short entry should be made in the "measures" field (e.g. done, completed, etc.) to make it easier to understand the evaluation results.

The total result of the QPN phase has to be documented on the result sheet (page 37). The relevant totals have to be determined and entered in the appropriate fields. The evaluation number has to be calculated, as shown in the example. In the software version of the QPN, this process is supported by the program.

Evaluation of the result sheet (Example: Product and process planning):

Evaluation	Evaluation phase				
	Product development			Series readiness	
	Concept	**Product-/ process-planning**	**Imple-mentati-on**	**Preparat-ion of 2-day proc.**	***Accept-ance 2DP*** [1]
Number of questions:		***15***			
No. Of questions not applicable:		***1***			
No. Of evaluated questions:		***14***			
Possible points:		***140***			
Points obtained:		***130***			
No. of quest. evaluated with "8"		***0***			
No. of quest. evaluated with "5"		***2***			
No. of quest. evaluated with "0"		***0***			
Readiness* : (Evaluation points: max. 10)		***9.3 ****			
Responsible for evaluation:					
Evaluation deadline:					

** **Readiness** = Points obtained / Possible points x 10*

[1] The acceptance, evaluation and documentation of 2-day production is performed with a specially designed protocol and is described in Chapter 4 "Series readiness".

The determined readiness (evaluation points) leads to the following conclusions:

< 8.5	=	Project status RED	=	Requirements not fulfilled, project goal at risk
> 8.5 - < 9.5	=	Project status YELLOW.	=	Requirements not completely fulfilled. Implementation of improvement measures mandatory.
> 9.5	=	Project status GREEN	=	Requirements almost fulfilled. Currently no problems expected.

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1.4 Determining the priority number for specific parts

The prioritisation of the parts is exclusively performed by the customer before the start of the Qualification Programme for New Parts, based on fixed evaluation criteria that are only described here for information purposes.

Parts with Priority 1: Important parts that require special monitoring
The project is jointly evaluated at the production site of the supplier in all QPN phases after coordination with the parts manager of the QA department of the customer plant.

Parts with Priority 2: Parts that require general monitoring
The supplier evaluates all QPN phases at his own responsibility, at the latest by the deadlines provided by the customer. The evaluation documents are presented to the parts manager of the QA department of the customer plant during an appointment.

Parts with Priority 3: Parts where monitoring is not required
The supplier evaluates all QPN phases at his own responsibility, at the latest by the project deadline specified by the customer. The evaluation documents are sent to the parts manager of the QA department of the customer plant.

Prioritisation block 1:	**Yes**	**No**
1.1 Will the supplier deliver this type of part to the Volkswagen Group for the first time?		
1.2 Is it a complex assembly?		
1.3 Is it a new design?		
1.4 Are new production technologies used (little or no experience by the supplier and/or customer)?		
1.5 Have equivalent/similar parts and/or the production site previously caused problems, e.g. production interruptions with Top-Q activities?		
1.6 Have equivalent/similar parts and/or the production site previously caused complaints because of field failures?		
1.7 Did the supplier have problems in the past with 2-day production compliance or did the evaluation deviate significantly from that of the customer?		
1.8 Have equivalent/similar parts led to significant problems at start-up in the past (experience of the parts manager of the QA department)?		
If no question is answered with Yes (x), then the part has Priority 3. If at least one question is answered with Yes (x) the part has Priority 2. In this case continue with Block 2.		

Prioritisation block 2:	**Yes**	**No**
2.1 Does the order concern a module without part No. / a JIT scope?		
2.2 Have equivalent/similar parts and/or the production site led to repeated sorting activities?		
2.3 Do equivalent/similar parts lead to an over-proportional number of assembly faults (0-km complaints)?		
2.4 Did equivalent/similar parts and/or the production site lead to severe complaints in the field / exceeding of fault prognosis (broken-down car, functional faults)?		
2.5 Are Q problems expected during the start-up due to the production technology, supplier combination or development competence?		
If at least one question in Block 2 is answered with Yes, the part has Priority 1. If no question is answered with Yes, the part retains Priority 2.		

Place	Date	Responsible OE manager QA	Person resp. for parts at QA dep.

1.5 Definition of terms and abbreviations

QA-personell and parts managers	This refers to the QA department responsible for this project in the customer plant that uses the component for the time in the VOLKSWAGEN GROUP. The department is represented by the parts manager who acts as a QA contact person in this programme.
Design sample release (BMG)	The design sample release is provided by the responsible development department. It requires approval by the technical department after previous testing of design samples produced with series tools and/or under conditions similar to series production and the required laboratory tests. The test sample approval is only valid as long as none of the characteristics (e.g. dimensions, material, manufacturer labels, production methods, production site, machinery, etc.) that were evaluated during the approval test are changed. The manufacturer is responsible for adhering to the approved manufacturing status.
PCI P_{pk} C_{pk}	Process Capability Index determined by Process Capability Analysis (see VDA 4, Part 1) Ppk = Before the start of the series Production (preliminary PCI) Cpk = After the start of the series Prodcution (long-term PCI)
MCI C_{mk}	Machine Capability Index (see VDA 4, Part 1) Short-term MCI that takes the position of the process into account.
Cubing	Representation of the body with all mounting elements to simulate the installation of components in the body under assembly conditions.
D-charac-teristic	Characteristic of a product that makes documentation mandatory
D-part	Part with a characteristic that makes documentation mandatory.
FMEA	Failure Mode and Effect Analysis. Analysis of potential failures and their consequences.
JIT	Just-In-Time (supply process synchronised with production requirements)
O-series	The purpose of a O-series is an evaluation before the application of a new product, that the parts and assemblies fit, are dimensionally to specification, and to check the function of the tools, test devices and installations under production conditions. It thus includes the complete production process as precursor to the series production and all parts manufactured have to be manufactured by tools used in the series production.
PVS	The Pre-production run is used for fine tuning of the production and assembly processes of the individual parts. The production has to be performed through processes that are typical for serial production and with tools that are used in series production.
RPS	Reference Point System
SOP	Start of Production
TL	Technical conditions of delivery
TLD sheet	Sheet/catalogue in which all characteristics are recorded for which documentation is mandatory.
VW standard	Technical standards in the VOLKSWAGEN GROUP (e.g.: VW 105 40 Manufacturer code for vehicle parts)
Important characteris-tics	Important and function-relevant product, process and test characteristics have to be defined through a System-FMEA for the product in interdisciplinary teams. Further important characteristics may, for example, result from the subsequent System-FMEA for the process. They may relate to legal, safety, design or process issues as well as to important customer-oriented aspects.

Phase A

2.1 Project information and deadlines

Planning of activities for the Qualification Programme for New Parts requires the exchange of information.

The software version of the Qualification Programme for New Parts (QPN) queries all project-specific and quality-relevant information and deadlines. The inputs are supported by appropriate processing information. All entries are automatically transferred by the programme to the appropriate forms, which makes subsequent work easier.

All deadlines have to be planned and entered, taking the specifications in Chapter 1.2 into account.

Example:

Project data:

Project	Priority	Part description	Reference part number			Index	Drawing date
VW350	3	Protective side rail	1K3	853	515		06.02.03

Further part numbers allocated to the parts family:	1K3	853	516		
	1K3	853	515	A	06.02.03

Quality-relevant deadlines in QPN:

	Phases C/D/E/F have evaluation deadlines that need to be planned in the Qualification Programme for New Parts	Planned		
		Deadl.*	Start*	End
	Coordination of important characteristics w. customer			
	Design FMEA			
	Process FMEA			
C	QPN evaluation "Planning phase" (product/process)	*		
	Deadline for tools order			
	Tools acceptance deadline at tool maker			
D	QPN evaluation “Implementation phase”	*		
E	QPN evaluation "Preparation for 2-day production"	*		
F	QPN evaluation "Acceptance of 2-day production"	*		
1)	Product-specific deadline, if required	*		

* In case of order scopes that were classified with Priority 1 or 2 by the customer, a detailed coordination of the deadlines takes place after the first project evaluation (Phase A/B) has been received.

1) Additional product-specific requirements (e.g. tests, releases) might have to be fulfilled for certain orders, depending on the scope of value addition. If required, this will be coordinated with the supplier within the framework of the qualification programme.

Phase A

A standardised "Quality framework schedule" (see table on Page 18) is compiled, based on the deadlines provided by the supplier and the planned quality-relevant measures, after the data have been received.

The schedule is used to visualise important project deadlines for the customer and to check them for agreement with the project goals / specifications.

Independently of the added value to be provided, the deadlines of the individual measures and the delivery dates that are in the "red area" of the quality framework schedule indicate a risk to a timeous project start and must therefore be immediately redefined. This must involve the Purchasing Department and the Technical Departments concerned, especially QA.

Measures that are evaluated as falling into the yellow area may lead to a deviation from the planned quality goals and have to be redefined within the framework of the Qualification Program for New Parts.

Phase A

2.2 Input frame for project information and deadlines

<u>(see the programme)</u>

2.3 Description of the planned production process

The quality-relevant process steps have to be listed on the overview sheet "Planned production process" (2.4) and supplemented with a short description concerning the required information, to aid the evaluation of the individual QPN phases.

This information is usually already available, e.g. through the development of the product development plan, the process flow diagram, the FMEAs, the control plan and the operating material and test device planning. In some cases the information still has to be determined in the planning phase.

Information that is not yet available has to be updated later.

The overview sheet "Planned production process" has the following purpose:

- Provide a first overview of the production process that is planned by the supplier and the planned measures for quality assurance.
- Make it possible to take existing experience of the customer with current series parts (similar parts / production processes) into account during the planning phase.
- Simplify the processing of the questions (checklist points) with regard to the different process steps in QPN.

The description of the production process is not evaluated.

Further processing steps:

Phase B "Questionnaire for the detailed evaluation of the concept" follows immediately after the description of the production process.

The result of the evaluation has to be sent to the QA department of the customer plant that uses the service for the first time. The processing instructions in the introduction have to be taken into account.

The next steps in the Qualification Programme for New Parts are the Phases C to E (questionnaires). These have to be processed and sent to the customer on time. If required, the updated information for Phase A can be attached.

The QPN can be closed with the successful acceptance of the 2-day production audit.

The QA department of the customer will contact the supplier should the delivery scope be reclassified by the customer to Priority 1 or 2.

Phase A

2.4 Input frame for "Planned production sequence" - (Input frame does not correspond in detail with the software version)

Serial number	*The "most important", quality-relevant process steps of the planned production process have to be listed. (In case of several versions, the process steps of the "parts family" with the highest complexity or the highest quality requirements has to be listed.)*		*Important process parameters / critical characteristic requirements that affect, for example, further processing, function, reliability, life span and appearance of the product.*		*Planned test / measurement devices and installations for series monitoring*	*100 % = 100% test A = Automated M = Manual V = Visual*	*Comment (Measures, deadlines, responsible persons)*
No	**Process Step** (incl. information concerning purchased parts/suppliers used)	**Machine / system**	**Important process parameters**	**Important characteristics** [1] / **Tolerance**	**Test/ / measurement devices**	**Test per-formed**	**Comment**
1							
2							
3							
4							
5							
6							
7							

[1] Characteristics for the process capability investigation / documentation that are planned according to VDA 4, have to be marked with "*". The quality assurance department of the customer reserves the right to changes.

Phase A

2.5 Quality framework schedule

QPN- Quality framework schedule (Standard evaluation dates for parts with Priority 3) **Revision date:**

Project:		Part description:		Responsible:	Name:
Priority:		Part number:		Project management, supplier:	
		Drawing date:		QA customer plant:	
Delivery No.:		Supplier:		Development (customer):	
DUNS-No.:		Place of production:		Purchasing (customer):	

Telephone No.:

B- Released

PVS < CW-YY> **0 series** **SOP** Status Comment:

Calendar weeks before SOP x x x

		Schedule CW-YY
1	Receipt of order (letter of commitment)	*
2	Coordination of important characteristics	*
3	Design FMEA	*
4	Completion of system FMEA process	*
5	QPN phase C: Evaluation planning phase	
6	Ordering of tools	*
7	Tool acceptance date (at toolmaker)	*
8	QPN phase D: Evaluation of implementation phase	
9	Design approval (BMG)	*
10	Presentation of First Sample parts at customer	*
11	Date for the provision of parts (PPD) for PVS	*
12	Presentation of First Sample parts (Note 1) at customer	*
13	Date for the provision of parts (PPD) for 0 series	*
14		*
15		*
16		*
17		*
18	QPN phase E: Requirements 2-day production audit	
19	QPN phase F: Acceptance 2-day production audit	

Key:

Start **X** Planned completion **O**

* **Depending on required value addition scope**

Project status i.o.: (green) Project goal at risk: (yellow) Project start-up at risk: (red)

14 - 17 : Possible additional, product-specific measures

Phase B - D Questionnaire

No.	Question / requirement	Evalu-ation	Deviation / problem	Measure	Sched./ Status	Respon-sible

3.1 Phase B Questionnaire concerning the concept phase
3.2 Phase C Questionnaire concerning the product and process planning phase
3.3 Phase D Questionnaire concerning the implementation phase

(see the programme)

Phase E Series readiness

4. Series readiness

Preparation and acceptance procedure for 2-day production (2DP)

A 2-day production audit (2DP) has to be performed for all purchased parts in the Volkswagen Group to prevent quality and capacity problems. This usually takes place in the time between the O-series and SOP (Start of Production).

The goal of 2-day production audit is that the supplier provides proof of:

- Process and quality performance of the complete production process under series conditions.
- The ability to produce the required volume of acceptable parts for the customer on time with the available staff and equipment capacity.
- The suitability of the packaging, storage and transport containers and dunnages provided, as well as of the transport equipment and route with respect to the quantitative requirements.

A **condition** for the acceptance procedure for the 2DP is production under series conditions at the supplier and sub-suppliers. In addition, all required development and planning work as well as the initial sample test and the release by the customer have to be completed.

Before the 2DP, the conditions for production have to be evaluated, independently of the priority of the batch of parts that is to be evaluated. This measure minimises the joint efforts required.

For this purpose, the questionnaire (checklist) concerning the acceptance procedure for the 2DP has to be applied during Phase "E", and the results have to be sent to the customer. During production, this questionnaire is one of the tools to check that the quality requirements are being adhered to.

For parts with Priority 1, the acceptance procedure for the 2DP takes place at the same time as the quality assurance of the customer. The questionnaire completed during Processing Phase E serves the customer as a base. If the questionnaire results sent to the customer and the results found during the acceptance procedure differ, a re-evaluation is required.

The framework conditions for the acceptance procedure for the 2DP depend on the following:

- Complexity and number of derivatives
- Required volume and time in the start-up phase of the customer
- Availability of tools and systems for different derivatives

For complex order scopes with a high number of derivatives, it might therefore be necessary to break the acceptance procedure for the 2DP down into different steps for the derivatives required. The necessary detailed planning is part of the handling of the Qualification Programme for New Parts.

Phase E Series readiness

The following basic requirements have to be taken into account when determining the production volume in Phase E:

1. The production volumes for parts with priorities 1 and 2 are set in agreement with the customer, based on the following points 2 to 6. The production volumes for parts with priority 3 are set by the supplier at his own responsibility, also taking the following points into account.

2. The number of derivatives to be produced has to be agreed upon between the Quality Assurance Department of the customer plant and the supplier, when modules have to be assembled or ranges have to be produced JIT.

3. Production under series conditions has to be performed for a period of at least 6 hours or until the required volume for the first two calendar weeks according to the start-up curve for SOP is reached (see logistics call "Start-up curve). Alternatively, a volume of up to 2% of the required volume for the first six months after SOP can be defined.

4. If the production process entailed several tools of the same kind or multiple templates, the 2DP has to be produced with at least 50% of the planned tool volume or at least 50% of the clusters of multiple templates.

5. For volumes that consist of several derivatives, the acceptance has to be performed for the derivative with the highest complexity or the highest quality requirements.

6. The timely supply of the established packaging and transport containers / dunnages in sufficient quantities has to be coordinated with the Logistics Department of the customer before acceptance.

The quality assurance departments of the individual customer plants may make different agreements in individual cases. These agreements have to be documented in the acceptance protocol.

The acceptance procedure for the 2DP is performed as follows:

- **For parts with Priority 1, always after coordination of the deadlines with the quality assurance department of the customer at the supplier's premises.**

- For parts with Priority 2 and 3, by the supplier at his own responsibility, as long as no other specifications were made by the customer during Phase E.

If the acceptance procedure for the 2DP results in **Status „RED“:** requirements not fulfilled, appropriate measures have to be agreed upon and implemented without delay. The acceptance procedure has to be repeated. If the result is **Status „YELLOW“:** requirements insufficiently / not completely fulfilled, the acceptance procedure has to be repeated if the deviations occurred for a technical reason. If the classification was based on other reasons, e.g. missing releases, a repeat of the 2DP is not necessary in each case.

If the relevant documents can be provided, the status is changed to "GREEN". The change of status has to be documented with a new acceptance protocol.

Phase E Series readiness

Overview of the phases to series readiness:

Phase	Evaluation phase	Presentation deadline at the customer	Result documents
E	Requirements for the acceptance procedure for the 2-day production (quick check)	After positive evaluation of the implementation phase, at the latest 1 week before the planned (agreed) acceptance deadline for the 2-day production	**(1) ; (2) ; (3) ;**
F	Acceptance procedure for the 2-day production (2DP)	After positive evaluation (Phase E) and initial sample test, at the latest 2 weeks before the Start of Production (SOP)	**(4)**

Result documents expected by the customer:

(1) : Quality framework plan (Phase A)
(2) : Description of the production sequence (Phase A)
(3) : Completed questionnaire (checklist) with evaluation result (result sheet)
(4) : Acceptance protocol for the 2-day production (Chapter 4.2)

The supplier may be subject to charges if his performance gives rise to complaints or makes it necessary to repeat the acceptance procedure for the 2-day production.

Phase E Questionnaire concerning acceptance procedure for the 2-day production

No.	Question / requirement	Evalu-ation	Deviation / problem	Measure	Sched./ Status	Respon-sible

4.1 Phase E Questionnaire (Checklist) concerning the acceptance procedure for the 2-day production

(Quick check for evaluation, detailed information for problems/deviations/measures only required if evaluation <10 points)

(see the programme)

4.2 Handling and Acceptance protocol for 2-day production

(see the programme)

5. QPN-Documentation of results

5.1 Result sheet

(see the programme)

6. Escalation procedure

Problem escalation and problem solution

If problems with the Qualification Programme for New Parts occur, they have to be clearly identified and efficiently solved.

The goal of the escalation procedure is the support of the process for the supplier and the customer through structured and joint actions.

The escalation procedure is further intended for using the resources for problem-solving efficiently and to preserve the respective interests of the customer and supplier.

The following events can be triggers for the escalation procedure:

- Deadlines are not being adhered to
- Measures agreed upon are not implemented
- The quality capability / performance of the sub-suppliers is not assured
- Progress is not apparent
- Agreements are not adhered to
- Deviations in the progress of the project cannot be solved on the present level of responsibility

The three-level escalation procedure:

Level 1 = Quality discussion

Level 2 = Project review at the supplier

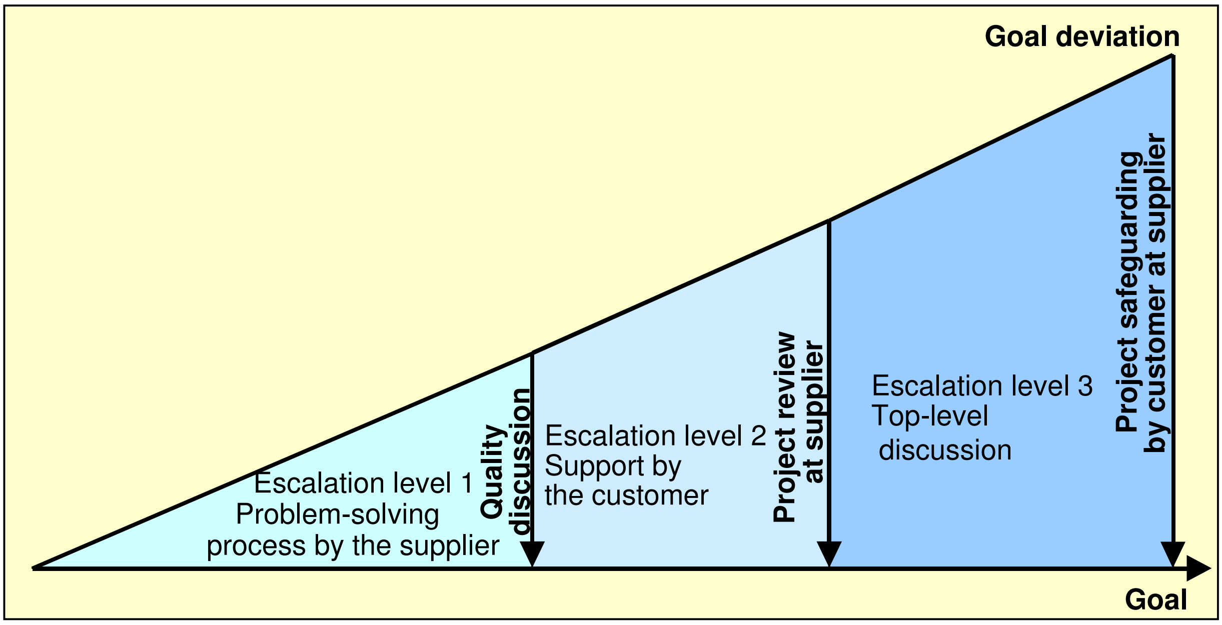
Level 3 = Project safeguarding by the customer plant at the supplier

Work content of every escalation level:

- Analysis of the reasons for escalation
- Analysis of the goal deviations / the problem
- Agreement on and implementation of appropriate measures
- Evaluation of the result
- Further escalation or conclusion of the process, depending on the evaluation

Escalation procuedure

The individual levels of escalation:



Escalation level 1 – Quality discussion

In a goal meeting, the supplier is asked about the goal deviation. The supplier should initiate and complete an effective problem-solving process to align the project with the goals. The participants are determined according to the type of problem (e.g. purchasing, development, pilot projects, quality assurance, logistics, production...)

Escalation level 2 – Project review at the supplier

Measures on Level 1 did not achieve the required results and it is therefore necessary to perform an analysis and track the measures at the supplier with the support of the customer.

Escalation level 3 – Project safeguarding by customer at the supplier's premises

Measures at previous levels did not achieve the required results and intensive involvement of the customer at the supplier's premises is therefore necessary. Top-level discussions between customer and supplier have to take place.